

ESAT PRACTICE TEST SERIES

PHYSICS + CHEMISTRY

LEVEL 1 • STANDARD

Physics + Chemistry

Mathematics 1 + Physics + Chemistry

Representative ESAT-style breadth with routine and multi-step applications.

MODULE

Mathematics 1

MODULE

Physics

MODULE

Chemistry

FULL SOLUTION BOOK

ThrivingScholars

81 preserved questions • three verified module keys

Physics + Chemistry • Level 1

Standard

This book compiles three existing 27-question modules into one 81-question pathway. No new questions have been generated and no source solution has been rewritten.

Composition: Mathematics 1 + Physics + Chemistry. Each module retains its original question order, answer and worked solution.

Mathematics 1

Engineering Mock Test 3 · preserved Mathematics 1 module

Q1 C	Q2 B	Q3 E	Q4 C	Q5 D
Q6 D	Q7 E	Q8 A	Q9 A	Q10 D
Q11 D	Q12 C	Q13 B	Q14 A	Q15 B
Q16 E	Q17 E	Q18 E	Q19 D	Q20 E
Q21 D	Q22 D	Q23 C	Q24 D	Q25 D
Q26 A	Q27 C			

Physics

Engineering Mock Test 3 · preserved Physics module

Q1 D	Q2 E	Q3 B	Q4 A	Q5 E
Q6 D	Q7 F	Q8 E	Q9 A	Q10 C
Q11 D	Q12 F	Q13 E	Q14 H	Q15 C
Q16 F	Q17 A	Q18 H	Q19 D	Q20 A
Q21 D	Q22 F	Q23 E	Q24 B	Q25 H
Q26 G	Q27 C			

Chemistry

Preserved Chemistry Standard source set

Q1 C	Q2 D	Q3 B	Q4 C	Q5 E
Q6 B	Q7 E	Q8 C	Q9 D	Q10 D
Q11 E	Q12 F	Q13 B	Q14 E	Q15 C
Q16 B	Q17 D	Q18 C	Q19 A	Q20 C
Q21 B	Q22 C	Q23 D	Q24 C	Q25 A
Q26 E	Q27 C			

Mathematics 1

27 QUESTIONS • COMPLETE SOLUTIONS

Engineering Mock Test 3 · preserved Mathematics 1 module. Question wording, option order, answer and solution method are unchanged.

M1-01

Mock Test 3 - Mathematics 1

Answer **C****QUESTION 1**

There are 8 consecutive integers. If the sum of the first three is 63, what is the largest one?

A 17**B** 25**C** 27**D** 30**E** 31**WORKED SOLUTION**

Let the first integer be n . Then $n + (n + 1) + (n + 2) = 63$, so $3n + 3 = 63$, giving $n = 20$. The largest of the eight integers is 27.

QUESTION 2

Joshua is 10 years old and his father is 42 years old. After how many years, his father's age is three times Joshua's age?

A 5

B 6

C 7

D 8

E 9

WORKED SOLUTION

After t years, $42 + t = 3(10 + t)$. Hence $42 + t = 30 + 3t$, so $t = 6$.

QUESTION 3

N is a 7-digit number consists of 2 or 3. There are more 2s than 3s in N . If N is divisible by both 3 and 4, what is the remainder of N when it is divided by 7?

A 1

B 2

C 3

D 4

E 5

WORKED SOLUTION

Let k be the number of 3s. The digit sum is $14 + k$, so divisibility by 3 gives $k = 1$. Divisibility by 4 forces the last two digits to be 32. Thus $N = 2222232$, and $N \equiv 5 \pmod{7}$.

M1-04

Mock Test 3 - Mathematics 1

Answer **C**

QUESTION 4

The preliminary round of a violin competition is divided into four rounds. According to the rule, the average score of a percipient of these four rounds must be not less than 96 points in order to enter the finals. Shaelyn's scores in the first three rounds are 95, 97, and 94. What is the minimum score that she needs to get on the fourth round in order to enter the finals? (Hope Cup Math Competition, China)

A 96

B 97

C 98

D 99

E 100

WORKED SOLUTION

The total score needed is $4 \times 96 = 384$. The first three scores total $95 + 97 + 94 = 286$, so the fourth score must be at least $384 - 286 = 98$.

M1-05

Mock Test 3 - Mathematics 1

Answer **D**

QUESTION 5

A haunted house has 8 windows. In how many ways can George the Ghost enter the house by one window and leave by a different window?

A 28

B 32

C 40

D 56

E 63

WORKED SOLUTION

There are 8 choices for the entry window and then 7 different choices for the exit window, giving $8 \times 7 = 56$.

M1-06

Mock Test 3 - Mathematics 1

Answer **D**

QUESTION 6

Jackie bought a bottle (30 oz) of 90% alcohol from pharmacy to make homemade disinfectant spray. How many ounces of water should he add in order to dilute it into 75% alcohol? (Hint: The amount of alcohol remains the same.)

A 3

B 4

C 5

D 6

E 7

WORKED SOLUTION

The amount of alcohol is $30 \times 0.90 = 27$ oz. If the final concentration is 75%, then $0.75T = 27$, so $T = 36$. Water added: $36 - 30 = 6$ oz.

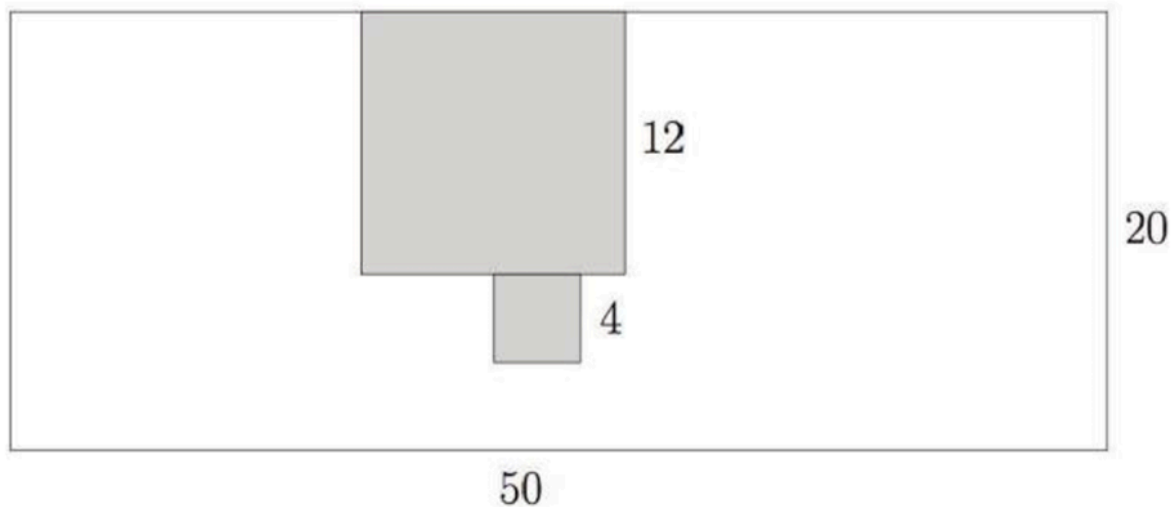
M1-07

Mock Test 3 - Mathematics 1

Answer E

QUESTION 7

Cut a 12×12 square and a 4×4 square from a 50×20 rectangle paper. What is the perimeter of the remaining part?



A 128

B 140

C 152

D 160

E 172

WORKED SOLUTION

The original rectangle perimeter is $2(50 + 20) = 140$. The 12×12 notch increases the perimeter by 24, and the attached 4×4 cut-out adds another 8. Total perimeter = $140 + 32 = 172$.

M1-08

Mock Test 3 - Mathematics 1

Answer A

QUESTION 8

Jasmine keeps adding numbers one by one from the following list

1, 3, 5, 7, ...

as $1 + 3 = 4$, $1 + 3 + 5 = 9$, $1 + 3 + 5 + 7 = 16$, etc. At one point, she got 379 and realized that one number is missed in the addition. What is the sum of digits of the missing number?

A 3

B 6

C 8

D 9

E 11

WORKED SOLUTION

The sum of the first n odd numbers is n^2 . Since 379 is just below $20^2 = 400$, the missing number is $400 - 379 = 21$. The digit sum is $2 + 1 = 3$.

M1-09

Mock Test 3 - Mathematics 1

Answer A

QUESTION 9

$\overline{ab}$ and $\overline{cd}$ are two 2-digit numbers. If

$$\overline{ab} - \overline{cd} = 24$$

and

$$\overline{1ab} - \overline{cd1} = 15,$$

what is $a + d$?

A 5

B 8

C 10

D 11

E 12

WORKED SOLUTION

Using place value, $\overline{ab} = 10a + b$ and $\overline{cd} = 10c + d$. Solving the two given equations gives $a + d = 5$.

M1-10

Mock Test 3 - Mathematics 1

Answer D

QUESTION 10

The speed of a train is 15 meters per second. If it takes the train 10 seconds to pass a lamp post, how long does it take the train to completely pass a 300-meter long tunnel, in seconds? (Hint: first figure out the length of the train based on the information related to the lamppost)

A 10

B 20

C 25

D 30

E 35

WORKED SOLUTION

The train length is $15 \times 10 = 150$ m. To pass the tunnel completely, it travels $300 + 150 = 450$ m. Time = $450 / 15 = 30$ s.

M1-11

Mock Test 3 - Mathematics 1

Answer D

QUESTION 11

How many pairs of positive integers (x, y) satisfy that $\frac{x}{3} = \frac{4}{y}$?

A 3

B 4

C 5

D 6

E 7

WORKED SOLUTION

From $x/3 = 4/y$, we get $xy = 12$. The positive factor pairs are $(1, 12)$, $(2, 6)$, $(3, 4)$, $(4, 3)$, $(6, 2)$, $(12, 1)$, so there are 6 pairs.

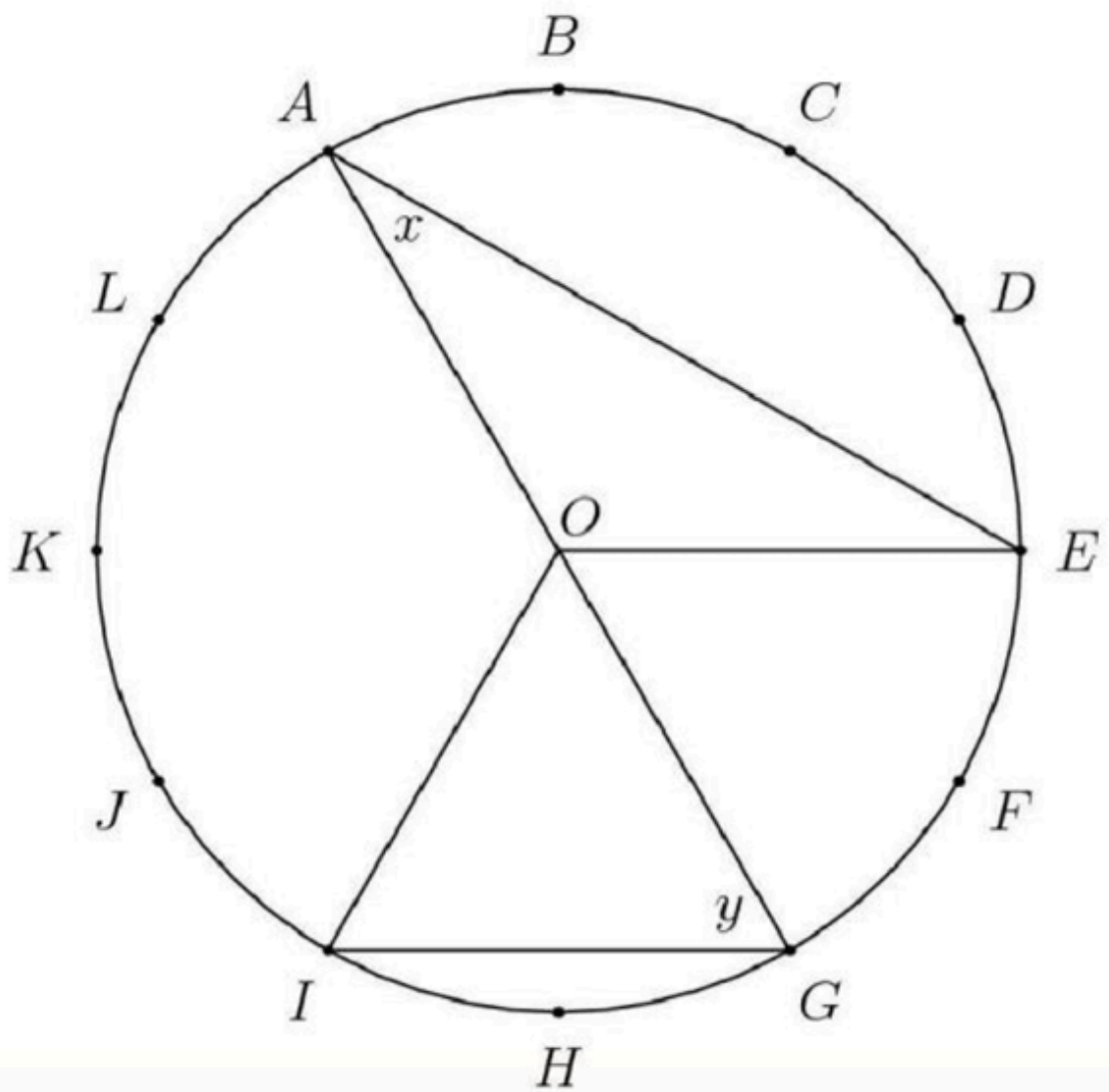
M1-12

Mock Test 3 - Mathematics 1

Answer C

QUESTION 12

The circumference of the circle with center O is divided into 12 equal arcs, marked the letters A through L as seen below. What is the number of degrees in the sum of the angles x and y ?



A 75

B 80

C 90

D 120

E 150

WORKED SOLUTION

The circle is divided into 12 equal arcs, so each arc is 30° . Using central and inscribed angle relationships from the diagram gives $x + y = 90^\circ$.

M1-13

Mock Test 3 - Mathematics 1

Answer **B**

QUESTION 13

If for whole numbers x , y and z ,

$$360 = 2^x \times 3^y \times 5^z,$$

what is $x + y + z$? Here $a^n = a \times a \times \dots \times a$.

For example, $100 = 2 \times 2 \times 5 \times 5 = 2^2 \times 5^2$.

A 5

B 6

C 7

D 8

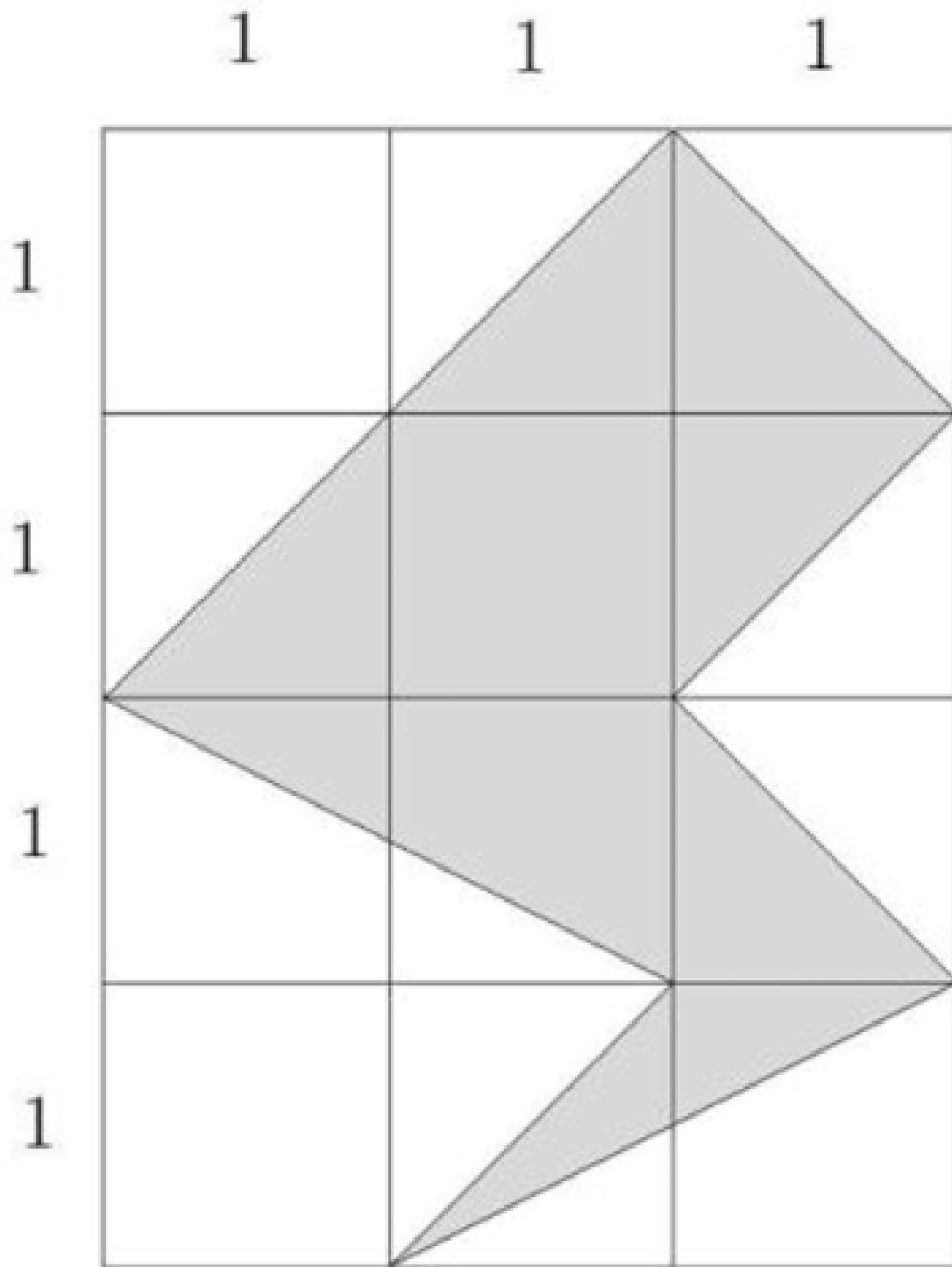
E 9

WORKED SOLUTION

Prime factorise: $360 = 36 \times 10 = 2^2 \times 3^2 \times 2 \times 5 = 2^3 \times 3^2 \times 5^1$. Hence $x + y + z = 3 + 2 + 1 = 6$.

QUESTION 14

What is the area of the shaded region in the following 3×4 grid?



A 5

B 6

C 7

D 8

E 9

WORKED SOLUTION

Counting the shaded polygon by subtracting the surrounding right triangles from the 3×4 rectangle gives shaded area 5.

M1-15

Mock Test 3 - Mathematics 1

Answer **B**

QUESTION 15

Randomly choose 3 different numbers from the set $\{1, 3, 5, 7, 9\}$. What is the probability that the three selected numbers could be side lengths of a triangle?

A $\frac{1}{5}$

B $\frac{3}{10}$

C $\frac{1}{3}$

D $\frac{2}{5}$

E $\frac{1}{2}$

WORKED SOLUTION

There are $\binom{5}{3} = 10$ triples. The valid triangle triples are $(3, 5, 7), (3, 7, 9), (5, 7, 9)$, so the probability is $3/10$.

M1-16

Mock Test 3 - Mathematics 1

Answer E

QUESTION 16

Marley practices exactly one sport each day of the week. She runs three days a week but never on two consecutive days. On Monday she plays basketball and two days later golf. She swims and plays tennis, but she never plays tennis the day after running or swimming. Which day of the week does Marley swim?

☐ A Sunday☐ B Tuesday☐ C Thursday☐ D Friday☒ E Saturday**WORKED SOLUTION**

Monday is basketball and Wednesday is golf. Applying the non-consecutive running rule and the condition that tennis is not immediately after running or swimming leaves Saturday as the swimming day.

QUESTION 17

Three toy cars run around a round track. It takes them 8, 10 and 12 seconds to finish one full circle respectively. If three cars start together from the same point and run toward the same direction, after how many seconds will they get back to the starting point together for the first time?

A 20

B 40

C 60

D 80

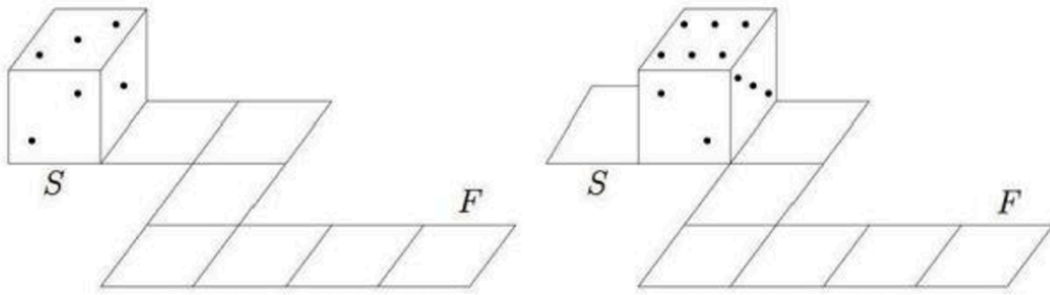
E 120

WORKED SOLUTION

The cars return together after the least common multiple of 8, 10, 12, which is 120 seconds.

QUESTION 18

Opposite faces of a die always add to 7. A die rolls on a circuit as represented below. At the starting point (S) the top face is 3. Which will be the top face at the final point (F)?



A 2

B 3

C 4

D 5

E 6

F 7

G 8

H 9

WORKED SOLUTION

Track the die through the path, using opposite faces summing to 7 after each roll.
The final top face is 6.

QUESTION 19

How many numbers x satisfy that $(x^2 - 2)^2 = 4$?

A 0

B 1

C 2

D 3

E 4

F 5

G 6

H 7

WORKED SOLUTION

$(x^2 - 2)^2 = 4$ gives $x^2 - 2 = \pm 2$. Thus $x^2 = 4$ or $x^2 = 0$, giving $x = -2, 0, 2$. There are 3 real numbers.

QUESTION 20

For how many two-digit numbers, if you switch the tens and ones digits (e.g., $18 \rightarrow 81$), the new 2-digit number is 18 more than the original one?

A 3

B 4

C 5

D 6

E 7

WORKED SOLUTION

Let the number be $10a + b$. Reversing digits gives $10b + a$. The condition gives $10b + a = 10a + b + 18$, so $b - a = 2$. There are 7 possible tens digits.

M1-21

Mock Test 3 - Mathematics 1

Answer D

QUESTION 21

[AMC 8 Problem] Malcolm wants to visit Isabella after school today and knows the street where she lives but doesn't know her house number. She tells him, "My house number has two digits, and exactly three of the following four statements about it are true."

- It is prime.
- It is even.
- It is divisible by 7.
- One of its digits is 9.

This information allows Malcolm to determine Isabella's house number. What is its units digit?

A 4

B 6

C 7

D 8

E 9

WORKED SOLUTION

The number must have exactly three of the four properties. The number 98 is even, divisible by 7, has a digit 9, and is not prime. Its units digit is 8.

M1-22

Mock Test 3 - Mathematics 1

Answer **D**

QUESTION 22

In the addition problem shown, m , n , p , and q represent positive digits. When the problem is completed correctly, the value of $m + n + p + q$ is (Canadian Math Competition)

t is prime.

t is even.

t is divisible by 7.

One of its digits is 9.

information allows N

A 16

B 18

C 22

D 24

E 30

WORKED SOLUTION

From the units column, $3 + 2 + q$ ends in 2, so $q = 7$ with carry 1. From the tens column, $6 + p + 8 + 1$ ends in 4, so $p = 9$. From the hundreds column,

$n + 7 + 5 + 2$ ends in 0, so $n = 6$ with carry $m = 2$. Therefore
 $m + n + p + q = 2 + 6 + 9 + 7 = 24$.

M1-23

Mock Test 3 - Mathematics 1

Answer C

QUESTION 23

If two numbers x and y satisfy that $(x + 1)^2 + (y - 5)^2 = 0$, what is $x + y$?

A -1

B 0

C 4

D 6

E 7

WORKED SOLUTION

A sum of squares is zero only if both squares are zero. Thus $x + 1 = 0$ and $y - 5 = 0$, so $x = -1$, $y = 5$, and $x + y = 4$.

M1-24

Mock Test 3 - Mathematics 1

Answer D

QUESTION 24

Abel tosses two coins and Ben tosses one coin. What is the probability that they get the equal number of heads?

Recall that

$$\text{Probability} = \frac{\text{Number of valid cases}}{\text{Number all possible cases}}.$$

A $\frac{1}{8}$

B $\frac{1}{4}$

C $\frac{1}{3}$

D $\frac{3}{8}$

E $\frac{2}{5}$

WORKED SOLUTION

There are 8 equally likely outcomes. Equal heads occurs when Abel gets 0 heads and Ben gets 0 heads, or when Abel gets 1 head and Ben gets 1 head:

$1 + 2 = 3$ cases. Probability = $3/8$.

M1-25

Mock Test 3 - Mathematics 1

Answer **D**

QUESTION 25

x , y , z and w are four numbers. If the average of the following three numbers

$x - y$, $y - z$, $z - w$ is 10 and $w = 5$. What is x ?

A 15

B 20

C 25

D 35

E there is no enough information

WORKED SOLUTION

The average of $x - y$, $y - z$, $z - w$ is 10, so their sum is 30. The sum telescopes: $(x - y) + (y - z) + (z - w) = x - w = 30$. Since $w = 5$, $x = 35$.

M1-26

Mock Test 3 - Mathematics 1

Answer A

QUESTION 26

What is the tens digit of 7^{2020} ?

A 0

B 1

C 2

D 3

E 4

WORKED SOLUTION

The last two digits of powers of 7 cycle: 07, 49, 43, 01. Since 2020 is divisible by 4, 7^{2020} ends in 01, so the tens digit is 0.

QUESTION 27

Two trains 300 miles apart are traveling toward each other along the same track. One train goes 70 miles per hour; the other runs at 80 miles per hour. If both of them leave their stations at 7:00am, when will they meet?

A 8 : 00 am

B 8 : 30 am

C 9 : 00 am

D 9 : 30 am

E 10 : 00 am

WORKED SOLUTION

The trains approach each other at $70 + 80 = 150$ miles per hour. Time
 $= 300 / 150 = 2$ hours, so they meet at 9:00 am.

Physics

27 QUESTIONS • COMPLETE SOLUTIONS

Engineering Mock Test 3 · preserved Physics module. Question wording, option order, answer and solution method are unchanged.

P-01

Mock Test 3 - Physics

Answer D

QUESTION 1

Which of the following statements are correct in relation to Newton's 3rd law?

- 1. For every action there is an equal and opposite reaction.
- 2. According to Newton's 3rd law, there are no isolated forces.
- 3. Rockets cannot accelerate in deep space because there is nothing to generate an equal and opposite force.

A Only 1**B** Only 2**C** Only 3**D** 1 and 2**E** 2 and 3**F** 1 and 3**WORKED SOLUTION**

Statement 1 is the common formulation of Newton's third law. Statement 2 presents a consequence of the application of Newton's third law.

Statement 3 is false: rockets can still accelerate because the products of burning fuel are ejected in the opposite direction from which the rocket needs to accelerate.

P-02

Mock Test 3 - Physics

Answer **E**

QUESTION 2

Which of the following statements are correct?

- 1. Positively charged objects have gained electrons.
- 2. Electrical charge in a circuit over a period of time can be calculated if the voltage and resistance are known.
- 3. Objects can be charged by friction.

A Only 1

B Only 2

C Only 3

D 1 and 2

E 2 and 3

F 1 and 3

G 1, 2 and 3

WORKED SOLUTION

Positively charged objects have lost electrons.

$$\text{Charge} = \text{Current} \times \text{Time} = \frac{\text{Voltage}}{\text{Resistance}} \times \text{Time}.$$

Objects can become charged by friction as electrons are transferred from one object to the other.

P-03

Mock Test 3 - Physics

Answer **B**

QUESTION 3

Which of the following statements is true?

- A** The gravitational force between two objects is independent of their mass.
- B** Each planet in the solar system exerts a gravitational force on the Earth.
- C** For satellites in a geostationary orbit, acceleration due to gravity is equal and opposite to the lift from engines.
- D** Two objects that are dropped from the Eiffel tower will always land on the ground at the same time if they have the same mass.
- E** All of the above.
- F** None of the above.

WORKED SOLUTION

Each body of mass exerts a gravitational force on another body with mass. This is true for all planets as well.

Gravitational force depends on the masses of both objects. Satellites stay in orbit due to centripetal force that acts tangentially to gravity (not because of the thrust from their engines). Two objects will only land at the same time if they also have the same shape or they are in a vacuum; otherwise air resistance gives different terminal velocities.

QUESTION 4

Which of the following best defines an electrical conductor?

- A** Conductors allow electrical charge to pass through them easily because they contain mobile charge carriers and have low resistance.
- B** Conductors are usually made from non-metals and they conduct electrical charge in multiple directions.
- C** Conductors are usually made from metals and they conduct electrical charge in one fixed direction.
- D** Conductors are usually made from non-metals and they conduct electrical charge in one fixed direction.
- E** Conductors allow the passage of electrical charge with zero resistance because they contain freely mobile charged particles.
- F** Conductors allow the passage of electrical charge with maximal resistance because they contain charged particles that are fixed and static.

WORKED SOLUTION

A conductor is a material that allows charge to move easily because it contains mobile charge carriers. Ordinary conductors have low resistance, not zero resistance.

QUESTION 5

An 800 kg compact car delivers 20% of its power output to its wheels. If the car has a mileage of 30 miles/gallon and travels at a speed of 60 miles/hour, how much power is delivered to the wheels? 1 gallon of petrol contains 9×10^8 J.

A 10 kW

B 20 kW

C 40 kW

D 50 kW

E 100 kW

WORKED SOLUTION

First, calculate the rate of petrol consumption:

$$\frac{\text{Speed}}{\text{Consumption}} = \frac{60 \text{ miles/ hour}}{30 \text{ miles/ gallon}} = 2 \text{ gallons/ hour}$$

Therefore, the total power is:

$$2 \text{ gallons} = 2 \times 9 \times 10^8 = 18 \times 10^8 \text{ J}$$

$$1 \text{ hour} = 60 \times 60 = 3600 \text{ s}$$

$$\text{Power} = \frac{\text{Energy}}{\text{Time}} = \frac{18 \times 10^8}{3600}$$

$$P = \frac{18}{36} \times 10^6 = 5 \times 10^5 \text{ W}$$

Since efficiency is 20%, the power delivered to the wheels is

$$5 \times 10^5 \times 0.2 = 10^5 \text{ W} = 100 \text{ kW}.$$

P-06

Mock Test 3 - Physics

Answer **D****QUESTION 6**

Which of the following statements about beta radiation are true?

- 1. After a beta particle is emitted, the atomic mass number is unchanged.
- 2. Beta radiation can penetrate paper but not aluminium foil.
- 3. A beta particle is emitted from the nucleus of the atom when an electron changes into a neutron.

A 1 only**B** 2 only**C** 1 and 3**D** 1 and 2**E** 2 and 3**F** 1, 2 and 3**WORKED SOLUTION**

Beta radiation is stopped by a few millimetres of aluminium, but not by paper. In β -radiation, a neutron changes into a proton plus an emitted electron, so the atomic mass number remains unchanged.

QUESTION 7

A car with a weight of 15,000 N is travelling at a speed of 15 m s^{-1} when it crashes into a wall and is brought to rest in 10 milliseconds. Calculate the average braking force exerted on the car by the wall. Take $g = 10 \text{ m s}^{-2}$.

A $1.25 \times 10^4 \text{ N}$

B $1.25 \times 10^5 \text{ N}$

C $1.25 \times 10^6 \text{ N}$

D $2.25 \times 10^4 \text{ N}$

E $2.25 \times 10^5 \text{ N}$

F $2.25 \times 10^6 \text{ N}$

WORKED SOLUTION

Firstly, calculate the mass of the car:

$$\text{mass} = \frac{\text{Weight}}{g} = \frac{15000}{10} = 1500 \text{ kg.}$$

Then using $v = u + at$ where $v = 0 \text{ m s}^{-1}$, $u = 15 \text{ m s}^{-1}$ and $t = 10 \times 10^{-3} \text{ s} = 0.01 \text{ s}$.

$$a = \frac{0 - 15}{0.01} = 1500 \text{ m s}^{-2}.$$

$$F = ma = 1500 \times 1500 = 2\,250\,000 \text{ N.}$$

QUESTION 8

Which of the following statements are correct?

- 1. Electrical insulators are usually metals e.g. copper.
- 2. The flow of charge through electrical insulators is extremely low.
- 3. Electrical insulators can be charged by rubbing them together.

A Only 1

B Only 2

C Only 3

D 1 and 2

E 2 and 3

F 1 and 3

G 1, 2 and 3

WORKED SOLUTION

Electrical insulators offer high resistance to the flow of charge. Insulators are usually non-metals; metals conduct charge very easily. Since charge does not flow easily to even out, they can be charged with friction.

QUESTION 9

The graph below represents a car's movement. At $t = 0$ the car's displacement was 0 m.

Which of the following statements are NOT true?

- 1. The car is reversing after $t = 30$.
- 2. The car moves with constant acceleration from $t = 0$ to $t = 10$.
- 3. The car moves with constant speed from $t = 10$ to $t = 30$.

The flow of charge through electrical insulators can be charged by rubbing.

Only 1

C. Only 3

Only 2

D. 1 and 2

The graph below represents a car's movement.

A 1 only

B 2 only

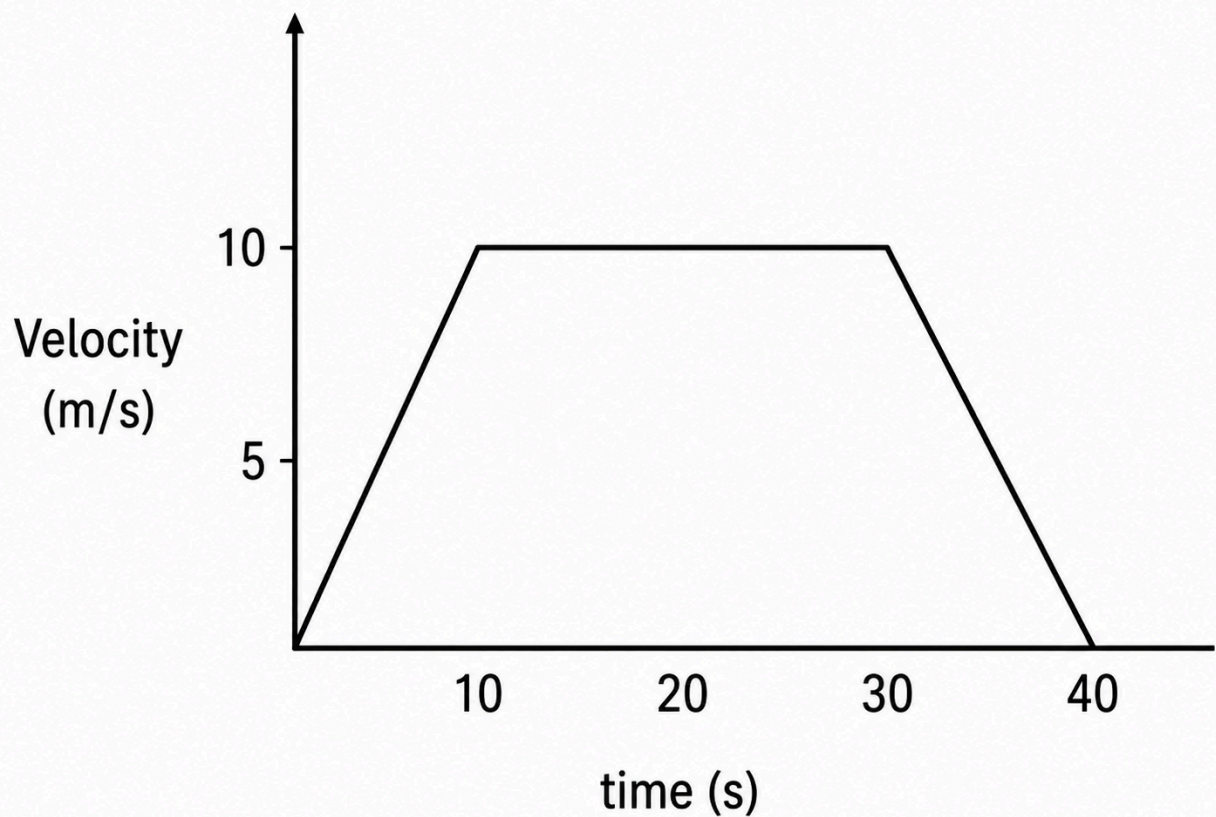
C 3 only

D 1 and 3

E 1 and 2

F 2 and 3

G 1, 2 and 3



WORKED SOLUTION

The car accelerates for the first 10 seconds at a constant rate and then decelerates after $t = 30$ seconds. It does not reverse, as the velocity is not negative. Therefore, only statement I is not true.

QUESTION 10

A 1,000 kg rocket is launched during a thunderstorm and reaches a constant velocity 30 seconds after launch. Suddenly, a strong gust of wind acts on it for 5 seconds with a force of 10,000 N in the direction of movement. What is the resulting change in velocity?

A 0.5 m s^{-1}

B 5 m s^{-1}

C 50 m s^{-1}

D 500 m s^{-1}

E 5000 m s^{-1}

F More information needed

WORKED SOLUTION

Using $F = ma$, where the unknown acceleration is $a = \frac{\Delta v}{\Delta t}$.

Hence: $\frac{F}{m} = \frac{\Delta v}{\Delta t}$.

Given $F = 10,000 \text{ N}$, $m = 1,000 \text{ kg}$ and $\Delta t = 5 \text{ s}$.

So $\frac{10,000}{1,000} = \frac{\Delta v}{5}$.

Therefore $\Delta v = 10 \times 5 = 50 \text{ m s}^{-1}$.

QUESTION 11

A 0.5 tonne crane lifts a 0.01 tonne wardrobe by 100 cm in 5,000 milliseconds. Calculate the average power developed by the crane. Take $g = 10 \text{ m s}^{-2}$.

A 0.2 W

B 2 W

C 5 W

D 20 W

E 50 W

F More information needed

WORKED SOLUTION

This question tests conversion to SI units and selection of relevant values.

0.01 tonnes = 10 kg; 100 cm = 1 m; 5,000 ms = 5 s.

$$P = \frac{\text{Work Done}}{t} = \frac{F \times d}{t}$$

The force is the weight of the wardrobe: $10 \times g = 10 \times 10 = 100 \text{ N}$.

$$\text{Thus, } P = \frac{100 \times 1}{5} = 20 \text{ W}.$$

QUESTION 12

A 20 V battery is connected to a circuit consisting of a 1 Ω and 2 Ω resistor in parallel. Calculate the overall current of the circuit.

A 6.67 A

B 8 A

C 10 A

D 12 A

E 20 A

F 30 A

WORKED SOLUTION

Recall the resistance of a parallel circuit: $\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$.

Thus, $\frac{1}{R_T} = 1 + \frac{1}{2} = \frac{3}{2}$ and therefore $R = \frac{2}{3} \Omega$.

Using Ohm's Law: $I = \frac{V}{R} = \frac{20 \text{ V}}{\frac{2}{3} \Omega} = 20 \times \frac{3}{2} = 30 \text{ A}$.

QUESTION 13

Which of the following statements is correct?

Assume the light ray crosses the boundary at a non-zero angle to the normal.

- A** The speed of light changes when it enters water.
- B** The speed of light changes when it leaves water.
- C** The direction of light changes when it enters water.
- D** The direction of light changes when it leaves water.
- E** All of the above.
- F** None of the above.

WORKED SOLUTION

At a non-zero angle to the normal, light changes speed when it passes between air and water and also changes direction due to refraction. Hence all four statements are true under the stated assumption.

P-14

Mock Test 3 - Physics

Answer H

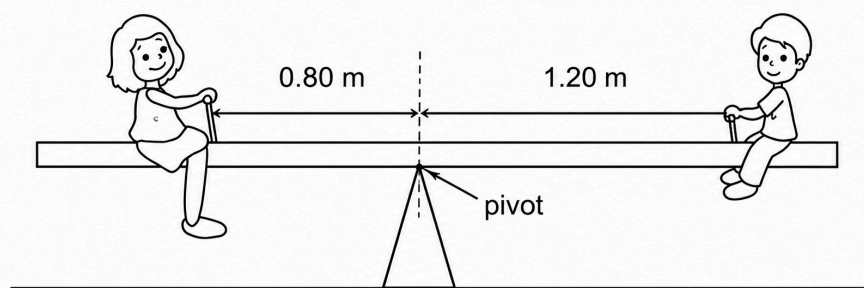
QUESTION 14

A plank of non-uniform density which has a mass of 15 kg is used to make a see-saw. A pivot is placed under the centre of the plank as shown on the diagram.

A boy of mass 35 kg sits at one end of the plank with his centre of gravity 1.20 m from the pivot. The see-saw balances when a woman of mass 60 kg sits on the plank on the other side of the pivot. Her centre of gravity is 0.80 m from the pivot.

Where is the centre of gravity of the plank and what is the magnitude of the force between the pivot and the plank?

(The gravitational field strength g is 10 N kg^{-1} .)



[diagram not to scale]

- A** 0.40 m on left of pivot; 100 N
- B** 0.40 m on left of pivot; 1100 N
- C** at the pivot; 100 N
- D** at the pivot; 1100 N
- E** 0.20 m on right of pivot; 100 N
- F** 0.20 m on right of pivot; 1100 N
- G** 0.40 m on right of pivot; 100 N
- H** 0.40 m on right of pivot; 1100 N

WORKED SOLUTION

Assume the centre of mass is at x metres to the right of the pivot.

Moments about pivot (CW positive):

$$35g(1.20) + Wx - 60g(0.8) = 0$$

$$Wx = 60g(0.8) - 35g(1.2)$$

$$Wx = 600(0.8) - 350(1.2) = 480 - 420 = 60$$

$$x = \frac{60}{W} = \frac{60}{150} = \frac{2}{5} = 0.40 \text{ m}$$

Newton's Second Law (vertical equilibrium on the plank):

$$R - W - 35g - 60g = 0$$

$$R = 150 + 350 + 600 = 1100 \text{ N}$$

P-15

Mock Test 3 - Physics

Answer C

QUESTION 15

A bath that is 1.0 m by 1.5 m is filled with water to a depth of 30 cm. The water in the bath is 40°C and it has been heated from the mains water at 20°C using a heating element that heats water on demand. It takes 3 minutes to run the bath and the heating element has an efficiency of 80%. The density of water, ρ , is 1000 kg m^{-3} , and the specific heat capacity, C_p , can be approximated as $4000 \text{ J kg}^{-1} \text{ K}^{-1}$.

What power is used by the heating element?

the bath is 40°C and it has been heated from the heating element that heats water on demand. It takes the heating element has an efficiency of 80%. The den and the specific heat capacity, C_p , can be approximate

Top

A 2500 kW

B 2000 kW

C 250 kW

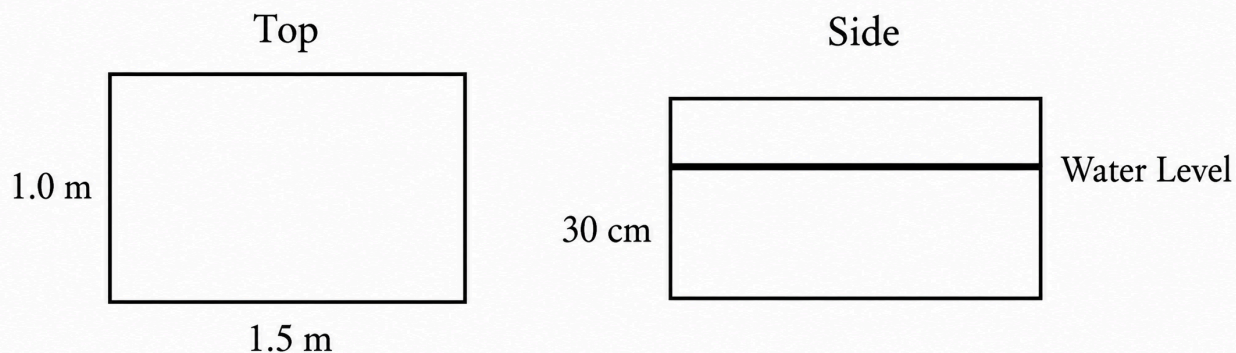
D 200 kW

E 100 kW

F 360 kW

G 3600 kW

H 1000 kW



WORKED SOLUTION

Answer: C

Work out how much energy is used to heat the water, then divide by the time taken to fill the bath to find the power used without accounting for losses. Then take account of the heating element efficiency to find the total power used.

Find the mass of water:

$$m = \rho V$$

$$V = 1 \times 1.5 \times 0.3 = 0.45 \text{ m}^3$$

$$\therefore m = 1000 \times 0.45 = 450 \text{ kg}$$

$$\text{Energy: } E = mc_p \Delta T$$

$$E = 450 \times 4000 \times 20 = 3.6 \times 10^7 \text{ J}$$

$$P = \frac{\Delta E}{\Delta t} = \frac{3.6 \times 10^7}{60 \times 3} = 2 \times 10^5 \text{ W} = 200 \text{ kW}$$

$$P = \text{Efficiency} \times P_{\text{tot}}$$

$$\therefore P_{\text{tot}} = \frac{P}{\text{Efficiency}} = \frac{200}{0.8} = 250 \text{ kW}$$

QUESTION 16

A 5 kW power source is used to generate a current through a circuit with a resistance of $100\ \Omega$. The charge of an electron, e , is $1.6 \times 10^{-19}\ \text{C}$. How many electrons pass through the circuit in 16 seconds?

A 4×10^{21}

B 5×10^{20}

C $4\sqrt{3} \times 10^{19}$

D 5×10^{18}

E $5\sqrt{2} \times 10^{21}$

F $5\sqrt{2} \times 10^{20}$

G $8\sqrt{2} \times 10^{22}$

H $8\sqrt{2} \times 10^{21}$

WORKED SOLUTION

Answer: F

The current is the flow of charge in the circuit. Work out the current and use it to find the total charge and consequently the total number of electrons that flow in a given time.

$$P = I^2 R$$

$$I^2 = \frac{P}{R} = \frac{5000}{100} = 50\ \text{A}^2$$

$$I = \sqrt{50} = 5\sqrt{2} \text{ A}$$

Remember where Q is charge: $I = \frac{\Delta Q}{\Delta t}$

$$\therefore \Delta Q = \Delta t \times I = 16 \times 5\sqrt{2} \text{ C}$$

$$\begin{aligned} \text{Number of electrons} &= \frac{\Delta Q}{e} \\ &= \frac{16 \times 5\sqrt{2}}{1.6 \times 10^{-19}} = \frac{16 \times 5\sqrt{2}}{16 \times 10^{-20}} = 5\sqrt{2} \times 10^{20} \end{aligned}$$

P-17

Mock Test 3 - Physics

Answer A

QUESTION 17

A shopper pushes a supermarket trolley a distance of 15 m in a straight line across a level, horizontal surface. The shopper applies a constant force of 50 N at an angle of 37° below the horizontal. The total weight of the trolley and its contents is 350 N.

[diagram not to scale]

What is the magnitude of the total vertical force that the surface exerts on the trolley and how much work is done by the pushing force?

(You may use the approximations $\sin 37^\circ = 0.60$; $\cos 37^\circ = 0.80$.)



A vertical force 380 N; work done 600 J

B vertical force 380 N; work done 750 J

C vertical force 390 N; work done 450 J

D vertical force 390 N; work done 750 J

WORKED SOLUTION

Question 17: A

Newtons Second law upwards on trolley:

$$R - 350 - 50 \sin(37) = 0$$

$$R - 350 - 50(0.6) = 0$$

$$R = 350 + 30 = 380 \text{ N}$$

Work done = Force $\times$ Distance:

$$W = 50 \cos(37) \times 15$$

$$W = 50(0.80) \times 15$$

$$W = 40 \times 15 = 600 \text{ J}$$

P-18

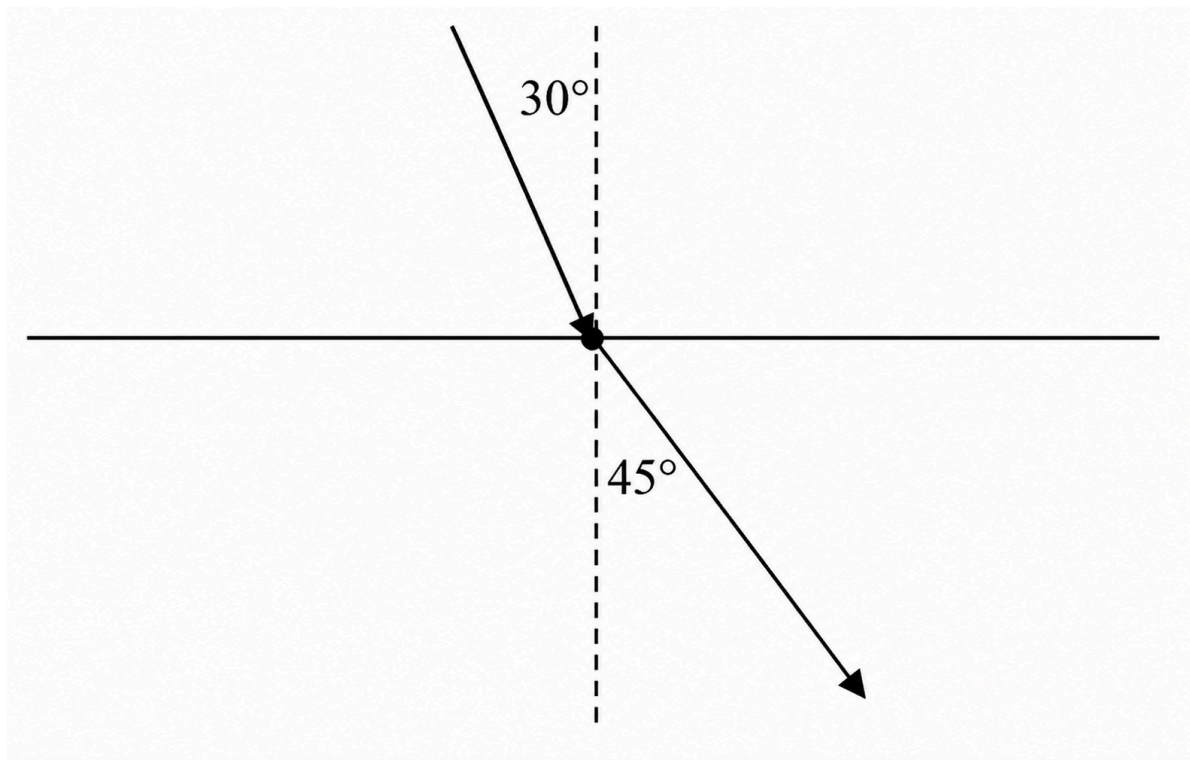
Mock Test 3 - Physics

Answer **H**

QUESTION 18

A wave approaches the boundary between two mediums at an angle of 30° to the normal. Upon entering the second medium it is refracted such that it now travels at 45° to the normal. The wave travels at 5 m s^{-1} in the first medium.

What velocity does the wave travel at in the second medium?



A $\frac{8}{\sqrt{3}} \text{ m s}^{-1}$

B $8\sqrt{3} \text{ m s}^{-1}$

C $\frac{8}{\sqrt{2}} \text{ m s}^{-1}$

D $10\sqrt{3} \text{ m s}^{-1}$

E $\frac{5}{2} \text{ m s}^{-1}$

F $10\sqrt{2} \text{ m s}^{-1}$

G $\frac{5}{\sqrt{2}} \text{ m s}^{-1}$

H $5\sqrt{2} \text{ m s}^{-1}$

WORKED SOLUTION

Question 18: H

$$n = \frac{v_1}{v_2} = \frac{\sin \theta_1}{\sin \theta_2}$$

$$n = \frac{\sin 30}{\sin 45}$$

$$\text{Remember: } \sin 30 = \frac{1}{2}, \sin 45 = \frac{1}{\sqrt{2}}$$

$$\therefore n = \frac{1}{2} \div \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$$

$$v_2 = \frac{v_1}{n} = 5 \div \frac{\sqrt{2}}{2} = \frac{10}{\sqrt{2}}$$

$$\therefore v_2 = 5\sqrt{2} \text{ m s}^{-1}$$

P-19

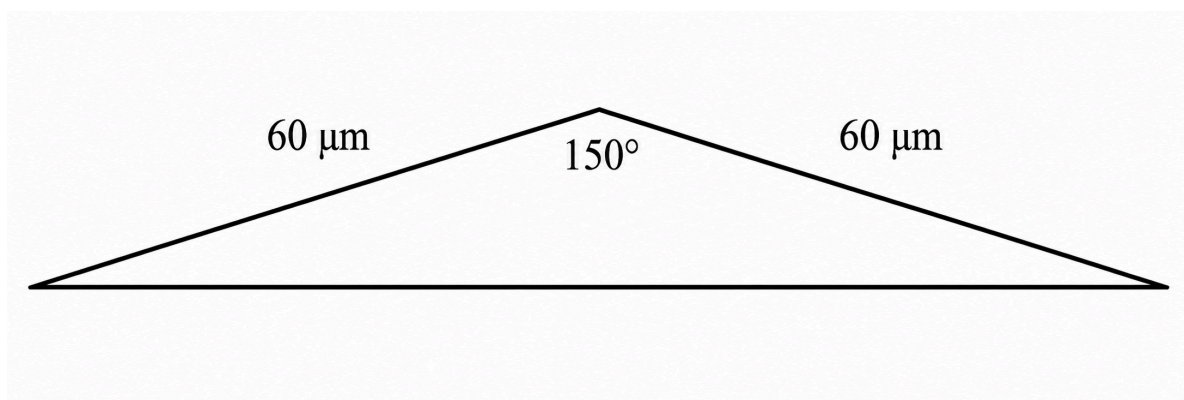
Mock Test 3 - Physics

Answer D

QUESTION 19

A strip of conductive material is plated onto a silicon surface to allow an electronic signal to be transmitted. A cross section of the plated material is given below, and the resistance is measured over a 5 cm length of the strip to be 1.0Ω .

What is the resistivity of the material?



A $6.0 \times 10^{-7} \Omega \text{m}$

B $3.6 \times 10^{-9} \Omega \text{m}$

C $2.4 \times 10^{-8} \Omega \text{m}$

D $1.8 \times 10^{-8} \Omega \text{m}$

E $2.0 \times 10^{-4} \Omega \text{m}$

F $3.6 \times 10^{-10} \Omega \text{m}$

G $5.4 \times 10^{-11} \Omega \text{m}$

H $4.2 \times 10^{-11} \Omega \text{m}$

WORKED SOLUTION

Work out the area of the cross section of the strip and use that to find the resistivity.

$$\text{resistivity, } \rho = \text{resistance} \times \frac{\text{cross sectional area}}{\text{length}} = \frac{RA}{l}$$

Recall area of triangle: $A = \frac{1}{2}ab \sin C$.

$$A = \frac{1}{2}(6 \times 10^{-5})^2 \sin(150^\circ)$$

Remember $\sin \theta$ is symmetrical about 90° so $\sin(150^\circ) = \sin(30^\circ) = \frac{1}{2}$.

$$\therefore A = \frac{1}{2}(6 \times 10^{-5})^2 \sin(30^\circ)$$

$$A = \frac{1}{2}(6 \times 10^{-5})^2 \times \frac{1}{2}$$

$$A = \frac{1}{4} \times 36 \times 10^{-10} = 9 \times 10^{-10} \text{ m}^2$$

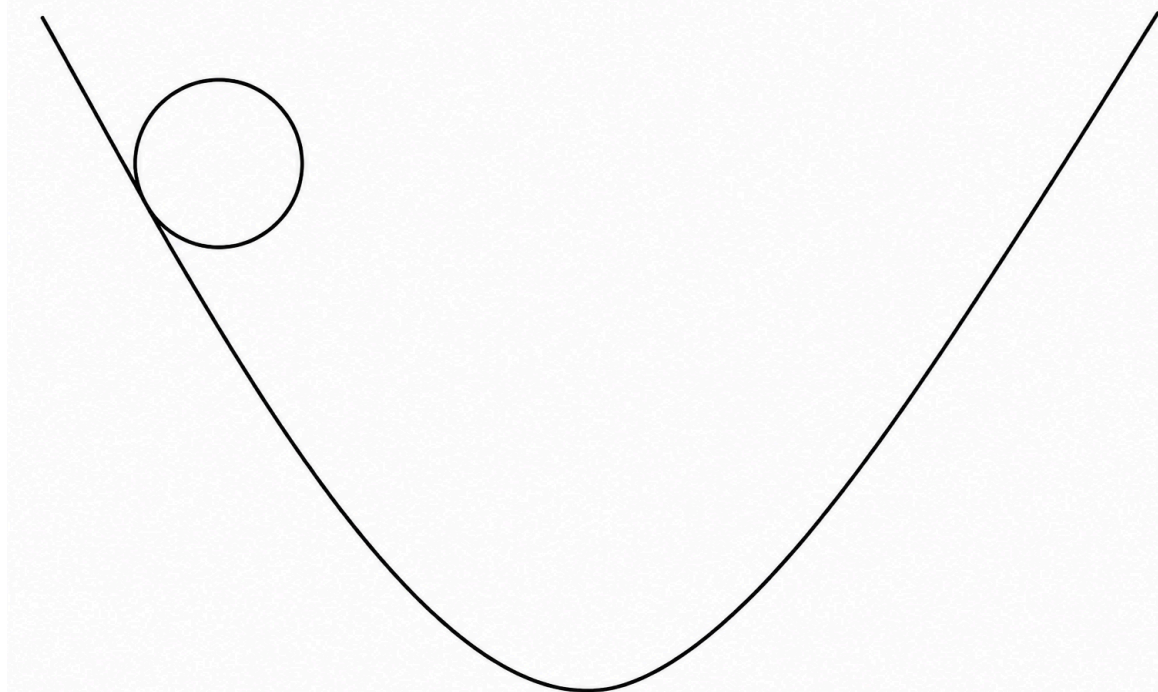
$$\rho = \frac{RA}{l} = \frac{1 \times 9 \times 10^{-10}}{0.05} = 9 \times 10^{-10} \times 20 = 1.8 \times 10^{-8} \Omega \text{m}$$

Note: This is approximately the resistivity of copper.

QUESTION 20

A ball rolls down and up a dip which can have its side profile modelled by the equation $y = 2x^2$. The point $(0, 0)$ corresponds to the bottom of the dip, and y is the height of the surface at horizontal distance x away from this centre.

What is the angle of the contact force acting on the ball when it is 3 m away from the lowest point horizontally? (Give the smallest angle between the direction of the force and the horizontal.)



A $\tan^{-1}\left(\frac{1}{12}\right)$

B $\tan^{-1}(12)$

C $\tan^{-1}(8)$

D $\tan^{-1}\left(\frac{1}{8}\right)$

E $\sin^{-1}\left(\frac{1}{12}\right)$

F $\sin^{-1}(12)$

G $\sin^{-1}(8)$

H $\sin^{-1}\left(\frac{1}{8}\right)$

WORKED SOLUTION

Question 20: A

You can evaluate the angle by looking at either $x = -3$ or $x = 3$ as the dip is symmetrical. Use differentiation to find the gradient at either point then use the gradient to work out the angle between the force and horizontal.

Find gradient at $x = 3$:

$$y = 2x^2$$

$$\frac{dy}{dx} = 4x$$

$$\Rightarrow \frac{dy}{dx} = 4 \times 3 = 12$$

Direction of the force is normal to the surface the ball is rolling on:

$$m' = -\frac{1}{m} = -\frac{1}{12}$$

Find the angle with the horizontal of this gradient:

$$\tan \theta = \frac{1}{12}$$

$$\theta = \tan^{-1}\left(\frac{1}{12}\right)$$

Note: This is approximately 5° but trig function is given as the paper is non-calculator.

QUESTION 21

A block of mass 2 kg sliding up a slope has an initial velocity of 4 m s^{-1} . The slope is angled at 30° to the horizontal and the block moves 1 m before it stops. (Take $g = 10 \text{ N kg}^{-1}$).

What is the average frictional force on the block as it moves up the slope?

A 2 N

B 1 N

C 3 N

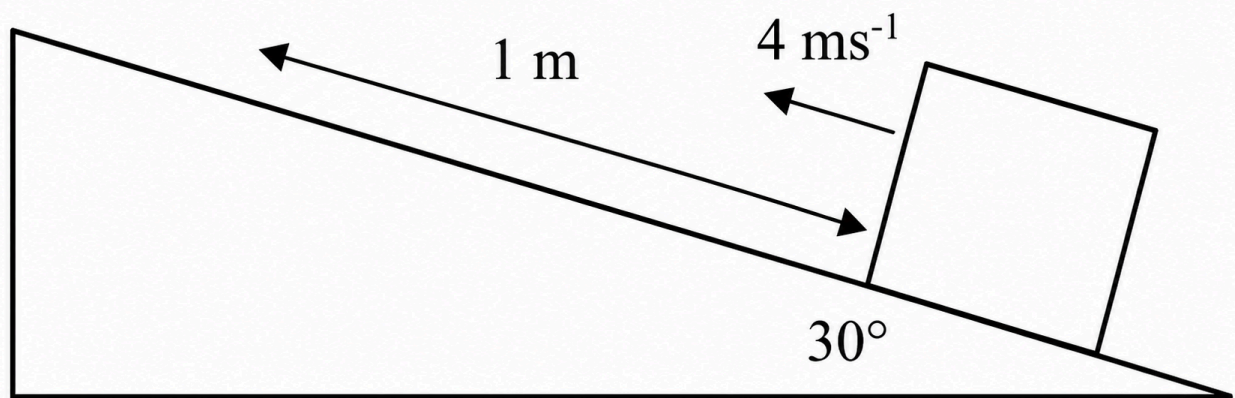
D 6 N

E 4 N

F 5 N

G 0 N

H 7 N



WORKED SOLUTION

Question 21: D

Do an energy balance for the conversion of kinetic energy into gravitational potential energy and work done against friction.

Change in height:

$$h = s \sin \theta = 1 \sin 30 = 10 \times \frac{1}{2} = 0.5 \text{ m}$$

Energy balance:

$$\text{KE} = \text{GPE} + \text{WD}$$

$$\frac{1}{2}mv^2 = mgh + F_f s$$

$$\frac{1}{2} \times 2 \times 4^2 = 2 \times 10 \times 0.5 + F_f \times 1$$

$$16 = 10 + F_f$$

$$F_f = 6 \text{ N}$$

QUESTION 22

An object is pushed with a force of 3 N for 5 s and then a second force of 4 N for 4 s. The forces act in the same direction and push an object of mass 5 kg. At the end of the 9 s period, what is the total change in momentum?

A 16 kg m s^{-1}

B 32 kg m s^{-1}

C 8 kg m s^{-1}

D 12 kg m s^{-1}

E 20 kg m s^{-1}

F 31 kg m s^{-1}

G 4 kg m s^{-1}

H 24 kg m s^{-1}

WORKED SOLUTION

The impulse equals the change in momentum. The two impulses act in the same direction, so

$$\Delta p = 3 \times 5 + 4 \times 4 = 15 + 16 = 31 \text{ kg m s}^{-1}.$$

QUESTION 23

Light travels at a speed of $2.5 \times 10^8 \text{ m s}^{-1}$ in a material. What is the critical angle for total internal reflection when light approaches an air boundary from within the material? (The speed of light in air is $3 \times 10^8 \text{ m s}^{-1}$).

A 30°

B 45°

C 60°

D $\sin^{-1}(\frac{4}{5})$

E $\sin^{-1}(\frac{5}{6})$

F $\sin^{-1}(\frac{3}{5})$

G $\sin^{-1}(\frac{2}{3})$

H $\sin^{-1}(\frac{4}{7})$

WORKED SOLUTION

Find the refractive index and then the critical angle.

$$n = \frac{c}{v}$$

$$n = \frac{3 \times 10^8}{2.5 \times 10^8} = \frac{6}{5} = 1.2$$

For the critical angle, $\sin C = \frac{1}{n}$.

$$\sin C = \frac{1}{1.2} = \frac{5}{6}$$

$$C = \sin^{-1}\left(\frac{5}{6}\right).$$

P-24

Mock Test 3 - Physics

Answer **B**

QUESTION 24

A spring is used to support a mass of 2 kg. The spring's spring constant, k , is 50 N m^{-1} . How much elastic potential energy is stored in the spring due to the mass it is supporting? (Take $g = 10 \text{ N kg}^{-1}$.)

A 1 J

B 4 J

C 20 J

D 25 J

E 8 J

F 15 J

G 14 J

H 30 J

WORKED SOLUTION

Work out the extension using Hooke's law and then the elastic potential energy.

$$W = F = kx$$

$$mg = kx$$

$$2 \times 10 = 50x \Rightarrow x = \frac{20}{50} = 0.4 \text{ m}$$

$$E = \frac{1}{2}Fx = \frac{1}{2}mgx$$

$$E = \frac{1}{2} \times 2 \times 10 \times 0.4 = 4 \text{ J}$$

P-25

Mock Test 3 - Physics

Answer **H**

QUESTION 25

A charge of 500 C is unloaded over a 30 s period. The voltage in the circuit is 90 V. What is the average power obtained?

A 600 W

B 450

C 250 W

D 300 W

E 100 W

F 1800 W

G 1000 W

H 1500 W

WORKED SOLUTION

Find the current from charge and time, then compute power.

$$I = \frac{\Delta Q}{\Delta t}$$

$$I = \frac{500}{30} = \frac{50}{3} \text{ A}$$

$$P = IV$$

$$\therefore P = \frac{50}{3} \times 90 = 1500 \text{ W}$$

P-26

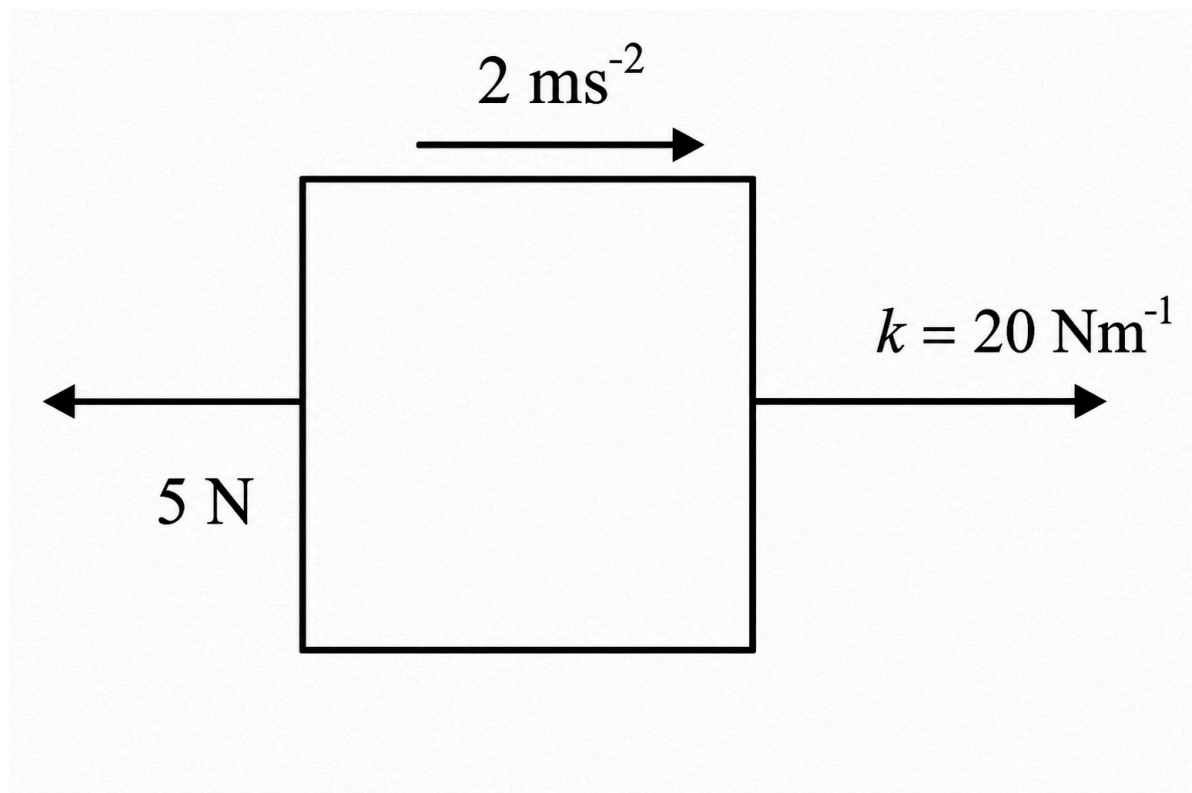
Mock Test 3 - Physics

Answer G

QUESTION 26

A block of mass 4 kg is pulled with a spring of spring constant, $k = 20 \text{ N m}^{-1}$. The acceleration of the block is 2 m s^{-2} and the force due to friction, F_f , is 5 N.

What is the extension in the spring?



A 22 cm

B 35 cm

C 8 cm

D 12 cm

E 5 cm

F 40 cm

G 65 cm

H 6 cm

WORKED SOLUTION

Resolve horizontally. The spring force must both overcome friction and accelerate the block:

$$F_s - 5 = 4 \times 2, \text{ so } F_s = 13 \text{ N.}$$

Using Hooke's law, $F_s = kx$, so $13 = 20x$. Hence $x = 0.65 \text{ m} = 65 \text{ cm}$.

P-27

Mock Test 3 - Physics

Answer C

QUESTION 27

A cylindrical wire is made of a material with a resistivity of $8 \times 10^{-8} \Omega \text{ m}$. It has a radius of 0.5 mm, and a length of 1 km. If the voltage in the wire is 400 V, what is the current?

A $\frac{10\pi}{3} \text{ A}$

B $\frac{2\pi}{5} \text{ A}$

C $\frac{5\pi}{4} \text{ A}$

D $\frac{6\pi}{25} \text{ A}$

E $\frac{25\pi}{3} \text{ A}$

F $\frac{13\pi}{4} \text{ A}$

G $\frac{17\pi}{4} \text{ A}$

H $\frac{11\pi}{5} \text{ A}$

WORKED SOLUTION

Question 27: C

Find the resistance using the resistivity and dimensions of the wire. Then solve with $V = IR$.

$$\text{resistivity, } \rho = \frac{RA}{l}$$

$$\therefore R = \frac{\rho l}{A}$$

$$A = \pi \times (0.5 \times 10^{-3})^2 = \frac{\pi}{4} \times 10^{-6} \text{ m}^2$$

$$R = \frac{8 \times 10^{-8} \times 1000}{\frac{\pi}{4} \times 10^{-6}} = \frac{320}{\pi} \Omega$$

$$I = \frac{V}{R}$$

$$\therefore I = 400 \div \frac{320}{\pi} = \frac{5\pi}{4} \text{ A}$$

Chemistry

27 QUESTIONS • COMPLETE SOLUTIONS

Preserved Chemistry Standard source set. Question wording, option order, answer and solution method are unchanged.

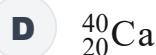
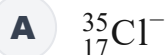
C-01

Atomic structure

Answer C

QUESTION 1

Which species has the **same number of neutrons and the same number of electrons** as ${}^{40}_{20}\text{Ca}^{2+}$?

**WORKED SOLUTION**

For ${}^{40}_{20}\text{Ca}^{2+}$, the number of neutrons is

$$40 - 20 = 20,$$

and the number of electrons is

$$20 - 2 = 18.$$

For ${}^{38}_{18}\text{Ar}$,

$$38 - 18 = 20 \text{ neutrons,} \quad 18 \text{ electrons.}$$

Both values match, so the correct answer is **C**.

C-02

Isotopes and relative atomic mass

Answer **D**

QUESTION 2

Element X occurs naturally as three isotopes of mass numbers 24, 25 and 26.

Their relative abundances are 6 : 1 : 1.

What is the relative atomic mass of X?

A 24.250

B 24.500

C 24.625

D 24.375

E 25.000

WORKED SOLUTION

Use the weighted mean:

$$A_r = \frac{6(24) + 1(25) + 1(26)}{6 + 1 + 1} = \frac{195}{8} = 24.375.$$

Therefore the correct answer is **D**.

QUESTION 3

Four atoms have the electron configurations shown.

atom	electron configuration
W	2,7
X	2,8,1
Y	2,8,7
Z	2,8,8,1

Which pair would be expected to react **most vigorously with each other** to form an ionic compound?

A W and X

B W and Z

C X and Y

D Y and Z

E W and Y

WORKED SOLUTION

W and Y are Group 17 atoms; X and Z are Group 1 atoms.

Group 1 reactivity increases down the group, so Z is more reactive than X. Group 17 reactivity decreases down the group, so W is more reactive than Y.

The most vigorous pairing is therefore the most reactive metal with the most reactive halogen: **W and Z**.

QUESTION 4

The properties of five substances are shown.

substance	melting point / °C	conducts when solid	conducts when molten
A	-20	no	no
B	660	good	good
C	805	no	good
D	1710	no	no
E	-114	no	no

Which substance is most likely to have a **giant ionic structure**?

A A

B B

C C

D D

E E

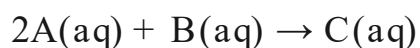
WORKED SOLUTION

A giant ionic solid normally has a high melting point. It does not conduct when solid because the ions cannot move, but it conducts when molten because the ions become mobile.

Only substance C has this combination of properties, so the answer is **C**.

QUESTION 5

The reaction



goes to completion.

Initially, $[\text{A}] = 0.80 \text{ mol dm}^{-3}$ and $[\text{B}] = 0.50 \text{ mol dm}^{-3}$, and no C is present. The volume does not change during the reaction.

What are the final concentrations of B and C?

A $[\text{B}] = 0.00, [\text{C}] = 0.50 \text{ mol dm}^{-3}$

B $[\text{B}] = 0.10, [\text{C}] = 0.20 \text{ mol dm}^{-3}$

C $[\text{B}] = 0.20, [\text{C}] = 0.40 \text{ mol dm}^{-3}$

D $[\text{B}] = 0.10, [\text{C}] = 0.80 \text{ mol dm}^{-3}$

E $[\text{B}] = 0.10, [\text{C}] = 0.40 \text{ mol dm}^{-3}$

WORKED SOLUTION

The reacting ratio is $2\text{A} : \text{B} : \text{C} = 2 : 1 : 1$.

Using all 0.80 mol dm^{-3} of A consumes

$$0.80 \div 2 = 0.40 \text{ mol dm}^{-3}$$

of B and forms the same concentration, 0.40 mol dm^{-3} , of C.

So

$$[\text{B}]_{\text{final}} = 0.50 - 0.40 = 0.10, \quad [\text{C}]_{\text{final}} = 0.40.$$

Therefore the answer is **E**.

C-06

Equilibrium

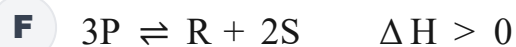
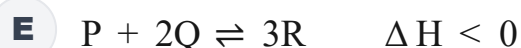
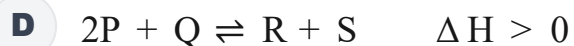
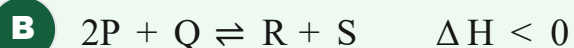
Answer **B**

QUESTION 6

For a reversible gaseous reaction:

- increasing the temperature **decreases** the equilibrium yield of R;
- increasing the pressure **increases** the equilibrium yield of R.

Which equation could represent the reaction?



WORKED SOLUTION

If increasing temperature decreases the product yield, the forward reaction must be exothermic, so $\Delta H < 0$.

If increasing pressure increases the product yield, the product side must contain fewer moles of gas.

In B there are 3 gaseous particles on the left and 2 on the right, and the forward reaction is exothermic. Therefore **B** is possible.

QUESTION 7

Chlorine reacts with cold aqueous hydroxide ions:



Consider the statements.

1. The oxidation state of chlorine in Cl_2 is 0.
2. The oxidation state of chlorine in ClO^- is +1.
3. Cl_2 acts only as an oxidising agent in this reaction.

Which statements are correct?

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

Elemental chlorine has oxidation state 0, so statement 1 is true.

In ClO^- , if chlorine has oxidation state x ,

$$x - 2 = -1 \Rightarrow x = +1,$$

so statement 2 is true.

Some chlorine is reduced from 0 to -1 and some is oxidised from 0 to $+1$.

Chlorine therefore acts as both an oxidising and a reducing agent, so statement 3 is false.

The answer is **E: 1 and 2 only**.

C-08

Chromatography

Answer C

QUESTION 8

Four dyes are separated using the same solvent. The solvent front moves 8.0 cm.

dye	distance moved / cm
W	2.0
X	3.6
Y	5.2
Z	6.4

In a second experiment, an unknown dye moves 3.25 cm while the solvent front moves 5.0 cm.

Which dye is the unknown most likely to be?

A W

B X

C Y

D Z

E It cannot be determined.

WORKED SOLUTION

For the unknown dye,

$$R_f = \frac{3.25}{5.0} = 0.65.$$

For dye Y,

$$R_f = \frac{5.2}{8.0} = 0.65.$$

So the unknown most likely corresponds to **Y**, answer **C**.

C-09

Empirical and molecular formulae

Answer **D**

QUESTION 9

A compound contains by mass:

- 40.0% carbon,
- 6.7% hydrogen,
- 53.3% oxygen.

Its relative molecular mass is 60.

$$A_r : \quad C = 12, H = 1, O = 16$$

What is its molecular formula?

A CH₂O

B C₂H₂O₂

C C₂H₄O

D C₂H₄O₂

E C₃H₆O₃

WORKED SOLUTION

For a 100 g sample, the amounts are approximately

$$\text{C} : \frac{40.0}{12} = 3.33, \quad \text{H} : \frac{6.7}{1} = 6.7, \quad \text{O} : \frac{53.3}{16} = 3.33.$$

Dividing by the smallest gives approximately 1 : 2 : 1, so the empirical formula is CH₂O.

Its empirical formula mass is 30. Since $60/30 = 2$, the molecular formula is



Therefore the answer is **D**.

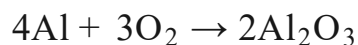
C-10

Limiting reagent and percentage yield

Answer **D**

QUESTION 10

Aluminium reacts with oxygen.



2.70 g of aluminium is mixed with 4.80 g of oxygen. The mass of aluminium oxide actually produced is 4.08 g.

$$A_r : \quad Al = 27, O = 16$$

What is the percentage yield?

A 40%

B 60%

C 75%

D 80%

E 85%

F 90%

WORKED SOLUTION

$$n(Al) = \frac{2.70}{27} = 0.100 \text{ mol}, \quad n(O_2) = \frac{4.80}{32} = 0.150 \text{ mol}.$$

Only 0.075 mol of O_2 is required for 0.100 mol of Al, so aluminium is limiting.

From the equation, 0.100 mol of Al can form 0.050 mol of Al_2O_3 .

$$M_r(Al_2O_3) = 102, \quad m_{\text{theoretical}} = 0.050(102) = 5.10 \text{ g}.$$

Thus

$$\% \text{ yield} = \frac{4.08}{5.10} \times 100 = 80\%.$$

The answer is **D**.

QUESTION 11

Solutions P and Q are both 0.10 mol dm^{-3} solutions of monoprotic acids.

- P is a strong acid.
- Q is a weak acid.

Consider the statements.

1. P has a lower pH than Q.
2. Equal volumes of P and Q require the same amount of sodium hydroxide for complete neutralisation.
3. Q has the greater electrical conductivity.

Which are correct?

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

The strong acid is more completely ionised, so it has the larger $[\text{H}^+]$ and hence the lower pH. Statement 1 is true.

Both acids are monoprotic and have the same formal concentration. Equal volumes therefore contain the same amount of acid and require the same amount of OH^- for complete neutralisation. Statement 2 is true.

The strong acid contains more mobile ions and therefore has the greater conductivity, so statement 3 is false.

The answer is **E: 1 and 2 only**.

C-12

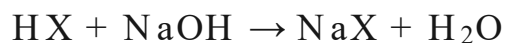
Titration

Answer **F**

QUESTION 12

A 1.56 g sample of a pure monoprotic acid HX is dissolved in water.

It requires exactly 25.0 cm^3 of 0.80 mol dm^{-3} sodium hydroxide for complete neutralisation.



What is the relative molecular mass of HX ?

A 39

B 52

C 65

D 72

E 104

F 78

WORKED SOLUTION

The amount of sodium hydroxide used is

$$n = CV = 0.80 \times 0.0250 = 0.0200 \text{ mol.}$$

The reaction ratio is 1:1, so the sample contains 0.0200 mol of HX.

Therefore

$$M_r(\text{HX}) = \frac{1.56}{0.0200} = 78.$$

The answer is **F**.

C-13

Rates of reaction

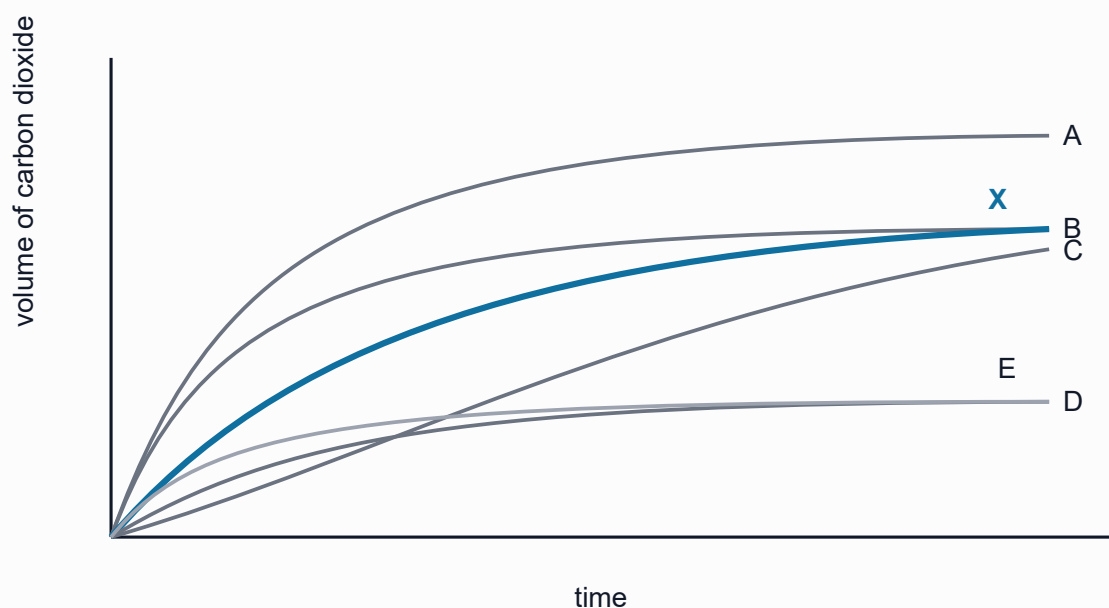
Answer **B**

QUESTION 13

Excess calcium carbonate reacts with 100 cm^3 of 1.0 mol dm^{-3} hydrochloric acid. Curve X shows the volume of carbon dioxide produced against time.

The experiment is repeated using 50 cm^3 of 2.0 mol dm^{-3} hydrochloric acid, with the same mass and particle size of excess calcium carbonate.

Which candidate curve represents the second experiment?



A curve A

B curve B

C curve C

D curve D

E curve E

WORKED SOLUTION

The amount of acid in the first experiment is

$$1.0 \times 0.100 = 0.100 \text{ mol.}$$

In the second experiment it is

$$2.0 \times 0.050 = 0.100 \text{ mol.}$$

So the same total amount of hydrochloric acid is available. Since calcium carbonate is in excess, the final amount of CO_2 is unchanged.

The second solution has twice the acid concentration, so the initial collision frequency and reaction rate are greater. The required curve must therefore reach the **same plateau more quickly**.

That is **curve B**.

C-14

Activation energy

Answer E

QUESTION 14

For a reaction,

$$\Delta H = -60 \text{ kJ mol}^{-1}.$$

The activation energy of the forward reaction is 90 kJ mol^{-1} .

What is the activation energy of the reverse reaction?

A -150 kJ mol^{-1}

B -30 kJ mol^{-1}

C 30 kJ mol^{-1}

D 90 kJ mol^{-1}

E 150 kJ mol^{-1}

F 210 kJ mol^{-1}

WORKED SOLUTION

The products are 60 kJ mol^{-1} below the reactants. The transition state is 90 kJ mol^{-1} above the reactants.

Therefore, measured from the products, the transition state is

$$90 + 60 = 150 \text{ kJ mol}^{-1}$$

higher. Hence the reverse activation energy is 150 kJ mol^{-1} , answer **E**.

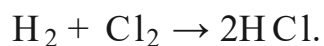
C-15

Bond energies

Answer **C**

QUESTION 15

Hydrogen reacts with chlorine:



Mean bond energies are:

bond	bond energy / kJ mol^{-1}
H-H	436
Cl-Cl	242
H-Cl	431

What is the approximate enthalpy change for the reaction?

A -678 kJ mol^{-1}

B -242 kJ mol^{-1}

C -184 kJ mol^{-1}

D $+184 \text{ kJ mol}^{-1}$

E $+678 \text{ kJ mol}^{-1}$

WORKED SOLUTION

Energy required to break the bonds in the reactants:

$$436 + 242 = 678 \text{ kJ mol}^{-1}.$$

Energy released when two H-Cl bonds form:

$$2(431) = 862 \text{ kJ mol}^{-1}.$$

Therefore

$$\Delta H = 678 - 862 = -184 \text{ kJ mol}^{-1}.$$

The answer is **C**.

QUESTION 16

Burning 0.050 mol of a fuel heats 100 g of water by 15°C.

The specific heat capacity of water is

$$4.2 \text{ J g}^{-1} \text{ }^{\circ}\text{C}^{-1}.$$

Only 70% of the energy released by the fuel is transferred to the water.

What is the enthalpy change of combustion of the fuel?

A -252 kJ mol^{-1}

B -180 kJ mol^{-1}

C -126 kJ mol^{-1}

D $+126 \text{ kJ mol}^{-1}$

E $+180 \text{ kJ mol}^{-1}$

F $+252 \text{ kJ mol}^{-1}$

WORKED SOLUTION

The energy absorbed by the water is

$$q = mc\Delta T = 100(4.2)(15) = 6300 \text{ J}.$$

This represents 70% of the energy released, so the fuel releases

$$\frac{6300}{0.70} = 9000 \text{ J} = 9.0 \text{ kJ}$$

when 0.050 mol burns.

Per mole, the magnitude is

$$\frac{9.0}{0.050} = 180 \text{ kJ mol}^{-1}.$$

Combustion is exothermic, so $\Delta H = -180 \text{ kJ mol}^{-1}$. The answer is **B**.

C-17

Electrolysis

Answer **D**

QUESTION 17

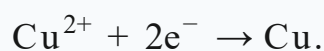
Inert electrodes are used.

Which row correctly gives the products?

- A** molten NaCl : cathode Na ; anode O_2
- B** dilute NaCl(aq) : cathode Na ; anode Cl_2
- C** $\text{CuSO}_4\text{(aq)}$: cathode H_2 ; anode Cu
- D** concentrated $\text{CuCl}_2\text{(aq)}$: cathode Cu ; anode Cl_2
- E** molten MgCl_2 : cathode H_2 ; anode Cl_2

WORKED SOLUTION

In concentrated aqueous copper(II) chloride with inert electrodes, Cu^{2+} is reduced at the cathode:



At the anode, chloride ions are oxidised:



So the correct row is **D**.

QUESTION 18

A 0.250 dm^3 solution initially contains 0.100 mol of Cu^{2+} .

Copper is electroplated from the solution until 3.20 g of copper has been deposited.

The volume is then restored to 0.250 dm^3 by adding water.

$$A_r(\text{Cu}) = 64$$

What is the final concentration of Cu^{2+} ?

A 0.10 mol dm^{-3}

B 0.15 mol dm^{-3}

C 0.20 mol dm^{-3}

D 0.25 mol dm^{-3}

E 0.40 mol dm^{-3}

WORKED SOLUTION

The amount of copper deposited is

$$n(\text{Cu}) = \frac{3.20}{64} = 0.0500 \text{ mol.}$$

Each mole of deposited copper removes one mole of Cu^{2+} , so the amount remaining is

$$0.100 - 0.0500 = 0.0500 \text{ mol.}$$

After the volume is restored to 0.250 dm^3 ,

$$[\text{Cu}^{2+}] = \frac{0.0500}{0.250} = 0.20 \text{ mol dm}^{-3}.$$

The answer is **C**.

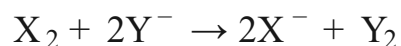
C-19

Halogen displacement

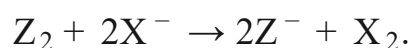
Answer **A**

QUESTION 19

The following reactions occur:



and



What is the order of reactivity of the halogens, from most to least reactive?

A $\text{Z} > \text{X} > \text{Y}$

B $\text{X} > \text{Z} > \text{Y}$

C $\text{X} > \text{Y} > \text{Z}$

D $\text{Y} > \text{X} > \text{Z}$

E $\text{Z} > \text{Y} > \text{X}$

WORKED SOLUTION

Because X_2 displaces Y^- , X is more reactive than Y:



Because Z_2 displaces X^- , Z is more reactive than X:



Therefore $\boxed{\text{Z} > \text{X} > \text{Y}}$, answer **A**.

QUESTION 20

Four metals M, Q, R, T give these observations.

1. Q displaces R from aqueous RSO_4 .
2. T also displaces R from aqueous RSO_4 .
3. T does not displace Q from aqueous QSO_4 .
4. R reacts with dilute hydrochloric acid.
5. M does not react with dilute hydrochloric acid.

Which is the correct order from most reactive to least reactive?

A $Q > R > T > M$

B $T > Q > R > M$

C $Q > T > R > M$

D $Q > T > M > R$

E $R > Q > T > M$

F $T > R > Q > M$

G $M > R > T > Q$

H $Q > M > T > R$

WORKED SOLUTION

Q displaces R, so $Q > R$. T displaces R, so $T > R$. T does not displace Q, so $Q > T$.

Thus $Q > T > R$.

R reacts with hydrochloric acid, so R is above hydrogen in the reactivity series. M does not react with hydrochloric acid, so M is below hydrogen and therefore below R.

The full order is $Q > T > R > M$, answer **C**.

C-21

Fractional distillation

Answer **B**

QUESTION 21

Hexane has a boiling point of 69°C . Octane has a boiling point of 126°C .

An approximately equal mixture of the liquids is separated using a fractionating column.

When the **first drops of distillate** are collected, which observation is most likely?

- A** flask below 69°C ; top near 69°C
- B** flask between 69°C and 126°C ; top near 69°C
- C** flask near 126°C ; top near 69°C
- D** flask between 69°C and 126°C ; top near 126°C
- E** flask near 126°C ; top near 126°C

WORKED SOLUTION

The liquid mixture in the flask boils over a temperature range lying between the boiling points of the two pure components.

Hexane is more volatile, so the vapour reaching the top of an efficient fractionating column is strongly enriched in hexane. When the first distillate is collected, the temperature near the top should therefore be close to 69°C .

Thus **B** is the most likely observation.

QUESTION 22

25 cm³ of a gaseous alkene is completely burned in excess oxygen.

100 cm³ of carbon dioxide is produced.

All gas volumes are measured at the same temperature and pressure.

Which is the molecular formula of the alkene?

A C₂H₄

B C₃H₆

C C₄H₈

D C₄H₁₀

E C₅H₁₀

WORKED SOLUTION

At the same temperature and pressure, gas volume ratios equal mole ratios.

The carbon dioxide volume is four times the alkene volume:

$$\frac{100}{25} = 4.$$

One molecule of C_nH_{2n} produces n molecules of CO₂, so n = 4.

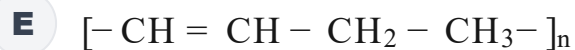
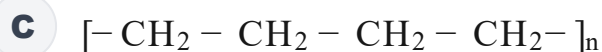
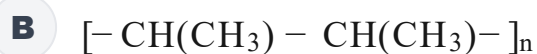
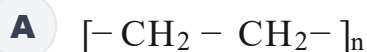
The alkene is C₄H₈, answer **C**.

QUESTION 23

But-1-ene has condensed structural formula



Which repeating unit represents poly(but-1-ene)?

**WORKED SOLUTION**

In addition polymerisation, the carbon–carbon double bond opens to form the polymer backbone. The ethyl group attached to the second alkene carbon remains as a side group.

The repeating unit is therefore



The answer is **D**.

QUESTION 24

A carboxylic acid X reacts with ethanol to form an ester Y and water.

The relative molecular mass of Y is 102.

$$M_r(\text{ethanol}) = 46, \quad M_r(\text{H}_2\text{O}) = 18.$$

What is the relative molecular mass of X?

A 46

B 60

C 74

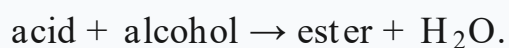
D 88

E 102

F 120

WORKED SOLUTION

Esterification is



Mass is conserved, so

$$M_r(\text{X}) + 46 = 102 + 18.$$

Hence

$$M_r(\text{X}) = 120 - 46 = 74.$$

The answer is **C**.

QUESTION 25

An aqueous solution of a salt gives:

- a **green flame**;
- a **blue precipitate** when aqueous sodium hydroxide is added;
- a **white precipitate** when acidified aqueous silver nitrate is added.

Which salt is present?

A CuCl_2

B CuSO_4

C FeCl_2

D NaCl

E CaCl_2

F $\text{Cu}(\text{NO}_3)_2$

WORKED SOLUTION

The green/blue-green flame and blue precipitate with sodium hydroxide indicate Cu^{2+} .

The white precipitate with acidified silver nitrate indicates chloride ions, because AgCl is white.

The salt is therefore $\boxed{\text{CuCl}_2}$, answer **A**.

QUESTION 26

Consider the statements.

1. Carbon monoxide can be formed by incomplete combustion of carbon-containing fuels.
2. Sulfur dioxide can be produced when sulfur-containing impurities in fuels are burned and can contribute to acid rain.
3. Nitrogen oxides are produced mainly because carbon burns with insufficient oxygen.

Which statements are correct?

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

Statement 1 is correct: incomplete combustion can produce carbon monoxide.

Statement 2 is correct: sulfur-containing impurities can form sulfur dioxide, which contributes to acid rain.

Statement 3 is false. Nitrogen oxides are formed when nitrogen and oxygen react at high temperatures, for example in combustion engines; they are not mainly a consequence of carbon burning with insufficient oxygen.

Therefore the answer is **E: 1 and 2 only**.

C-27

Quantitative chemistry

Answer C

QUESTION 27

A 6.00 g mixture contains only magnesium and magnesium oxide.

The mixture reacts completely with excess dilute hydrochloric acid.

Exactly 2.40 dm^3 of hydrogen gas is produced at room temperature and pressure.

At rtp,

$$1 \text{ mol gas} = 24.0 \text{ dm}^3.$$

$$A_r(\text{Mg}) = 24$$

What percentage by mass of the original mixture was magnesium?

A 20%

B 30%

C 40%

D 50%

E 60%

WORKED SOLUTION

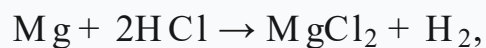
Only magnesium metal produces hydrogen with dilute hydrochloric acid.

Magnesium oxide neutralises the acid but does not form hydrogen.

The amount of hydrogen formed is

$$n(\text{H}_2) = \frac{2.40}{24.0} = 0.100 \text{ mol.}$$

From



the amount of magnesium is also 0.100 mol. Its mass is

$$0.100(24) = 2.40 \text{ g.}$$

Therefore

$$\% \text{Mg} = \frac{2.40}{6.00} \times 100 = 40\%.$$

The answer is **C**.

Compilation record: Level 1 uses Engineering Mock Test 3 for Mathematics/Physics and the corresponding Standard Biology/Chemistry source sets. This solution book is ready to support the later construction of the matching full test.